

Mach-Zehnder Interferometer

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Abstract—This paper presents the design process of a Mach-Zehnder Interferometer (MZI) from design to manufacturing.

Index Terms—Mach-Zehnder Interferometer, Lumerical

I. INTRODUCTION

Silicon photonics enables the integration of optical components onto a silicon platform using fabrication processes compatible with modern CMOS technology. Among the most widely used silicon photonic devices is the Mach-Zehnder interferometer (MZI), which exploits the phase difference between two optical paths to produce wavelength-dependent interference. Owing to its simple structure and high sensitivity to optical phase, the MZI is widely used in optical communication systems, sensing, filtering, and modulation applications.

In this work, a silicon photonic MZI was designed, simulated, fabricated, and experimentally characterized. The waveguide properties were first analyzed using Lumerical MODE to determine the effective and group indices, and the complete interferometer was simulated in Lumerical INTERCONNECT. The effects of fabrication tolerances were investigated through corner analysis. Experimental measurements obtained from the fabricated device were then processed using MATLAB to perform baseline correction, determine the free spectral range (FSR), and extract the group index from the measured transmission spectrum. The measured results were compared with the simulation and theoretical predictions to evaluate the performance of the fabricated device.

II. THEORY

A. Waveguide

For a lossless waveguide with length L and electric field E , the propagation equation of plane wave is:

$$E_0 = E e^{i\beta L} \quad (1)$$

where β is the propagation constant of light and is defined as:

$$\beta = \frac{2\pi n_{\text{eff}}}{\lambda} \quad (2)$$

where n_{eff} is the effective refractive index and λ is the wavelength.

B. Y-Branch Splitter/Combiner

Let's look at this ideal Y-Branch splitter. As can be seen from the image, in this Y-branch, for an incident beam with input intensity of I_i and corresponding electric field E_i , I_i and E_i is split equally between the two branches, resulting in $I_1 = I_2 = I_i/2$ and $E_1 = E_2 = E_i/\sqrt{2}$, respectively.

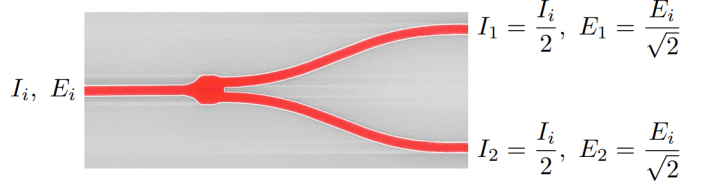


Fig. 1. Ideal Y-Branch splitter.

For an ideal Y-Branch combiner, as can be seen from the image, equation for the electric field at the output of the combiner is $E_o = (E_1 + E_2)/\sqrt{2}$

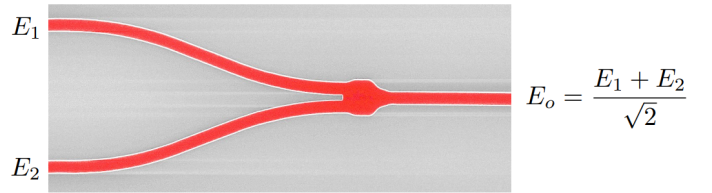


Fig. 2. Ideal Y-Branch combiner.

C. Mach-Zehnder Interferometer

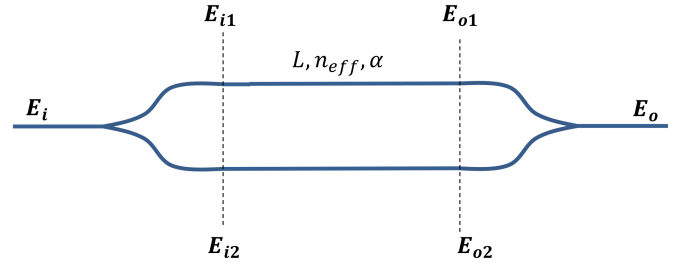


Fig. 3. Balanced MZI.

Mach-Zehnder Interferometer (MZI) is a 2 port device composed of a splitter and a combiner. In the splitter, light gets split into two arms and for the lossless case where the waveguide lengths are L_1 and $L_2 = L_1 + \Delta L$, from the equations above the output of each arm is given by

$$E_{o1} = E_{i1} e^{i\beta_1 L_1} = \frac{E_i}{\sqrt{2}} e^{i\beta_1 L_1} \quad (3)$$

$$E_{o2} = E_{i2} e^{i\beta_2 (L_1 + \Delta L)} = \frac{E_i}{\sqrt{2}} e^{i\beta_2 (L_1 + \Delta L)} \quad (4)$$

which results in

$$E_o = \frac{E_{o1} + E_{o2}}{\sqrt{2}} = \frac{E_i}{\sqrt{2}} (e^{i\beta_1 L_1} + e^{i\beta_2 (L_1 + \Delta L)}) \quad (5)$$

D. Mach-Zehnder Interferometer

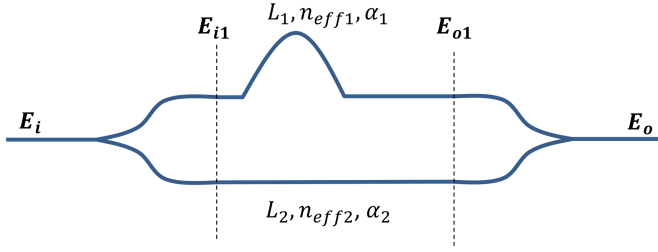


Fig. 4. Imbalanced MZI.

from which we can find the intensity of the output as:

$$I_o = \frac{I_i}{4} |e^{i\beta_1 L_1} + e^{i\beta_2(L_1 + \Delta L)}|^2 \quad (6)$$

which can be written as:

$$I_o = \frac{I_i}{2} [1 + \cos(\beta_1 L_1 - \beta_2 L_2)] \quad (7)$$

and since the two waveguides are identical, it can be written as

$$I_o = \frac{I_i}{2} [1 + \cos(\beta \Delta L)] \quad (8)$$

The free spectral range (FSR), which is the distance between adjacent peaks of the MZI transfer function is calculated as:

$$FSR = \frac{\lambda^2}{n_g(\lambda) \Delta L} \quad (9)$$

where group index (n_g) is defined as:

$$n_g(\lambda) = n_{eff}(\lambda) - \lambda \frac{dn_{eff}(\lambda)}{d\lambda} \quad (10)$$

III. MODELLING AND SIMULATIONS

A. Waveguide

This MZI design implements a strip waveguide with a width of 500 nm and a height of 220 nm and is designed for TE mode propagation. The TE and TM mode profiles, along with the effective index and group index as functions of wavelength, were simulated in Lumerical MODE and are presented below.

A waveguide compact model was also derived by fitting a Taylor expansion around the centre wavelength using a MATLAB script with the equation to find the Taylor coefficients:

$$n_{eff}(\lambda) = n_1 - n_2(\lambda - \lambda_0) - n_3(\lambda - \lambda_0)^2 \quad (11)$$

The script uses effective index data exported from Lumerical MODE as a function of wavelength.

$$n_{eff}(\lambda) = 2.447 - 1.133(\lambda - 1.55) - 0.044(\lambda - 1.55)^2 \quad (12)$$

From the above equations, effective index can be calculated as 2.447 and group index can be calculated as 4.203, which is similar to the values for effective index and group index we found through Lumerical MODE.

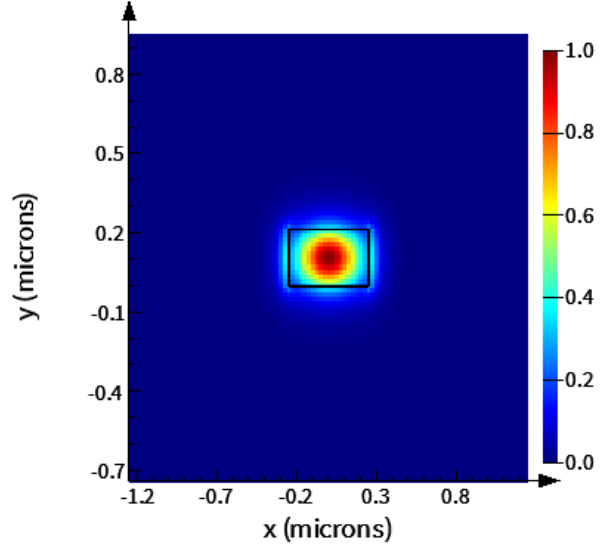


Fig. 5. Electric field intensity of TE_0 mode in the waveguide.

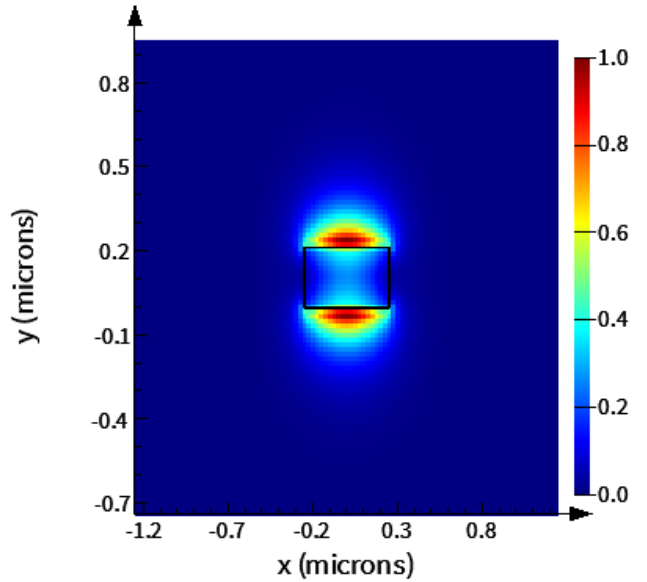


Fig. 6. Electric field intensity of TM_0 mode in the waveguide.

B. Layout and Mach-Zehnder Interferometer Design

The layout has 6 MZI designs. The first 5 MZIs are designed with y-splitters in both input and output with path differences of 200, 100, 50, 13 and 50 nm respectively. MZI6 was implemented using y-splitter at input and directional coupler at the output with a path difference of 46.25 nm.

The table below calculates FSR values for MZIs with various path length differences in the layout for TE_0 mode:

This is the gain versus wavelength plot of MZI1, which has a 200 nm path difference. The simulation was done with

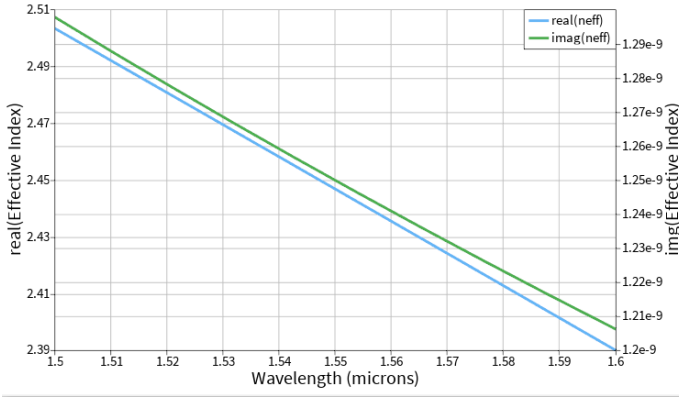


Fig. 7. Effective index versus wavelength.

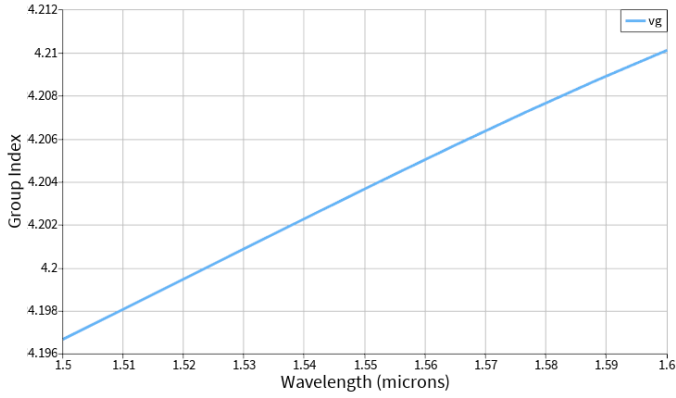


Fig. 8. Group index versus wavelength.

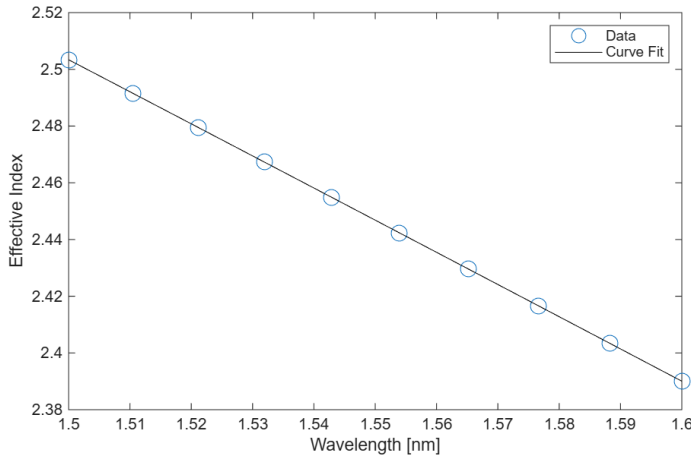


Fig. 9. Waveguide compact model for TE_0 .

TABLE I
COMPARISON OF FSR VALUES FOR DIFFERENT PATH LENGTH DIFFERENCES.

ΔL (μm)	FSR (nm)
13	43.97
46.25	12.36
50	11.43
100	5.72
200	2.86

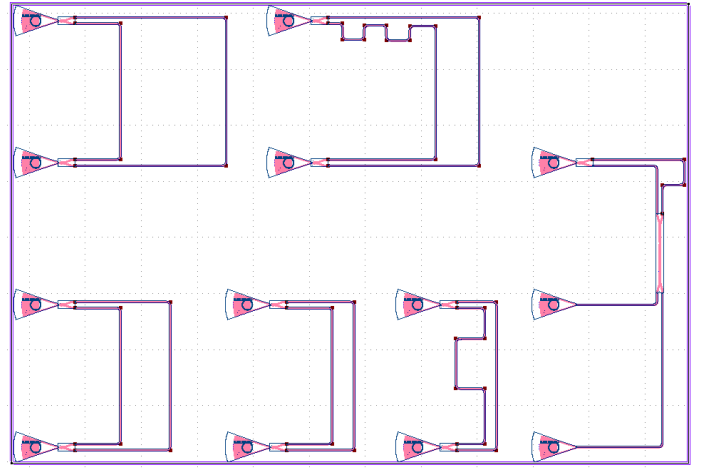


Fig. 10. Layout of 6 MZIs.

Lumerical INTERCONNECT.

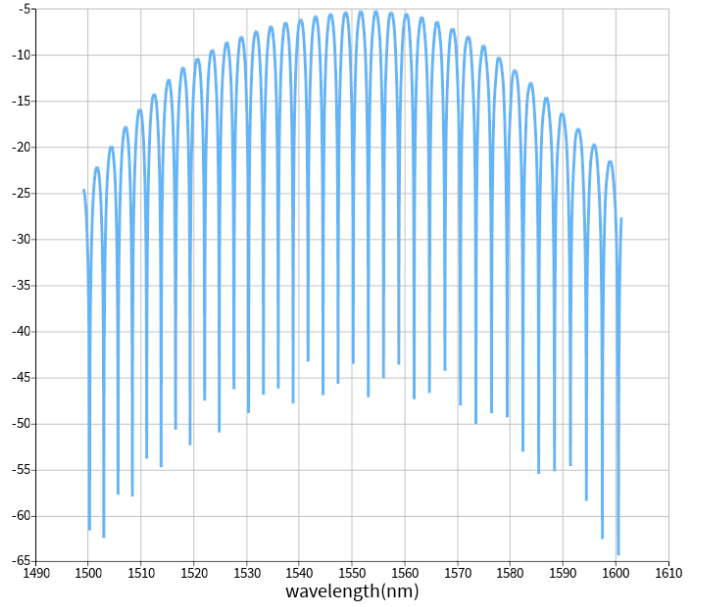


Fig. 11. Simulated gain-wavelength plot of MZI1

IV. FABRICATION

A. Process

1) *Applied Nanotools, Inc. NanoSOI process:* The photonic devices were fabricated using the NanoSOI MPW fabrication process by Applied Nanotools Inc. (<http://www.appliednt.com/nanosoi>; Edmonton, Canada) which is based on direct-write 100 keV electron beam lithography technology. Silicon-on-insulator wafers of 200 mm diameter, 220 nm device thickness and 2 μm buffer oxide thickness are used as the base material for the fabrication. The wafer was pre-diced into square substrates with dimensions of 25x25 mm, and lines were scribed into the substrate

backsides to facilitate easy separation into smaller chips once fabrication was complete. After an initial wafer clean using piranha solution (3:1 H₂SO₄:H₂O₂) for 15 minutes and water/IPA rinse, hydrogen silsesquioxane (HSQ) resist was spin-coated onto the substrate and heated to evaporate the solvent. The photonic devices were patterned using a JEOL JBX-8100FS electron beam instrument at The University of British Columbia. The exposure dosage of the design was corrected for proximity effects that result from the backscatter of electrons from exposure of nearby features. Shape writing order was optimized for efficient patterning and minimal beam drift. After the e-beam exposure and subsequent development with a tetramethylammonium sulfate (TMAH) solution, the devices were inspected optically for residues and/or defects. The chips were then mounted on a 4" handle wafer and underwent an anisotropic ICP-RIE etch process using chlorine after qualification of the etch rate. The resist was removed from the surface of the devices using a 10:1 buffer oxide wet etch, and the devices were inspected using a scanning electron microscope (SEM) to verify patterning and etch quality. A 2.2 μm oxide cladding was deposited using a plasma-enhanced chemical vapour deposition (PECVD) process based on tetraethyl orthosilicate (TEOS) at 300°C. Reflectometry measurements were performed throughout the process to verify the device layer, buffer oxide and cladding thicknesses before delivery.

2) *Washington Nanofabrication Facility (WNF) silicon photonics process:* The devices were fabricated using 100 keV Electron Beam Lithography [1]. The fabrication used silicon-on-insulator wafer with 220 nm thick silicon on 3 μm thick silicon dioxide. The substrates were 25 mm squares diced from 150 mm wafers. After a solvent rinse and hot-plate dehydration bake, hydrogen silsesquioxane resist (HSQ, Dow-Corning XP-1541-006) was spin-coated at 4000 rpm, then hotplate baked at 80 °C for 4 minutes. Electron beam lithography was performed using a JEOL JBX-6300FS system operated at 100 keV energy, 8 nA beam current, and 500 μm exposure field size. The machine grid used for shape placement was 1 nm, while the beam stepping grid, the spacing between dwell points during the shape writing, was 6 nm. An exposure dose of 2800 $\mu\text{C}/\text{cm}^2$ was used. The resist was developed by immersion in 25% tetramethylammonium hydroxide for 4 minutes, followed by a flowing deionized water rinse for 60 s, an isopropanol rinse for 10 s, and then blown dry with nitrogen. The silicon was removed from unexposed areas using inductively coupled plasma etching in an Oxford Plasmalab System 100, with a chlorine gas flow of 20 sccm, pressure of 12 mT, ICP power of 800 W, bias power of 40 W, and a platen temperature of 0 °C, resulting in a bias voltage of 185 V. During etching, chips were mounted on a 100 mm silicon carrier wafer using perfluoropolyether vacuum oil. Cladding oxide was deposited using plasma enhanced chemical vapor deposition (PECVD) in an Oxford Plasmalab System 100 with a silane (SiH₄) flow of 13.0 sccm, nitrous oxide (N₂O) flow of 1000.0 sccm, high-purity nitrogen (N₂) flow of 500.0 sccm, pressure at 1400mT, high-frequency RF power of 120W, and

a platen temperature of 350C. During deposition, chips rest directly on a silicon carrier wafer and are buffered by silicon pieces on all sides to aid uniformity.

B. Corner Analysis

In this section, manufacturing variability of the MZI was used. Firstly, corner analysis was ran for the strip waveguide wavelength with widths of 470, 510 nm and lengths of 215.3 and 223.1 nm compared to the nominal values of 500 nm width and 220 nm height. Here are the corner analysis results for group and effective index.

TABLE II
RESULTS FOR THE NOMINAL WAVEGUIDE AND PROCESS CORNERS.

Case	Width (nm)	Thickness (nm)	Group Index	Effective Index
Nominal	500	220.0	4.204	2.450
Corner 1	470	215.3	4.253	2.372
Corner 2	470	223.1	4.262	2.406
Corner 3	510	215.3	4.180	2.442
Corner 4	510	223.1	4.188	2.472

Next, for $\Delta L = 100\mu\text{m}$, corner analysis was ran to find FSR.

TABLE III
CALCULATED FREE SPECTRAL RANGE (FSR) FOR A PATH LENGTH DIFFERENCE OF 200 μm AT A WAVELENGTH OF 1550 NM.

Case	Group Index	FSR (nm)
Nominal	4.2036	2.858
Corner 1	4.2530	2.824
Corner 2	4.2620	2.819
Corner 3	4.1800	2.874
Corner 4	4.1880	2.869

V. MEASUREMENTS

To characterize the devices, a custom-built automated test setup [2, 6] with automated control software written in Python was used [3]. An Agilent 81600B tunable laser was used as the input source and Agilent 81635A optical power sensors as the output detectors. The wavelength was swept from 1500 to 1600 nm in 10 pm steps. A polarization maintaining (PM) fibre was used to maintain the polarization state of the light, to couple the TE polarization into the grating couplers [4]. A 90° rotation was used to inject light into the TM grating couplers [4]. A polarization maintaining fibre array was used to couple light in/out of the chip [5].

The data taken was measured at 30 °C for a TE polarization with a wavelength sweep around 1550 nm. Results for all 6 MZIs in conditions specified above are shown below:

VI. ANALYSIS

The aim in this part is getting the waveguide parameters for the measured case for MZI1 where $\Delta L = 200\mu\text{m}$. This consists of several steps first of which is using baseline correction method given in the course, centered around 1530-1570 nm:

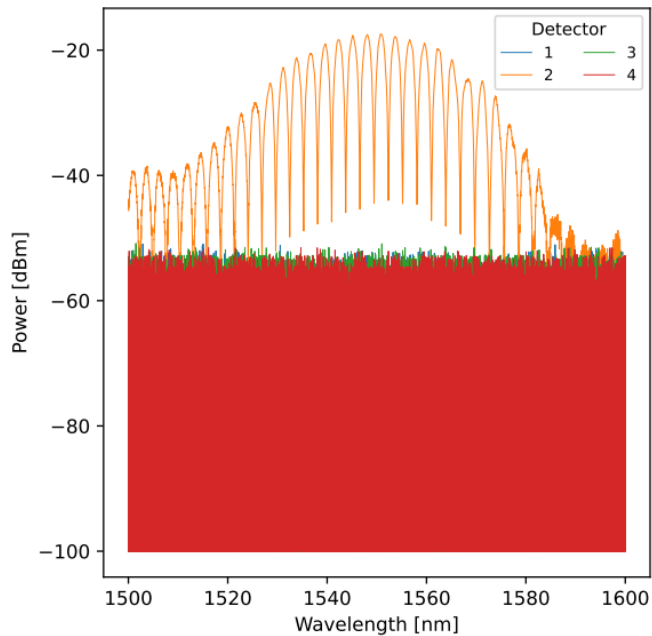


Fig. 12. Measured spectrum of MZI1 where $\Delta L = 200\mu m$

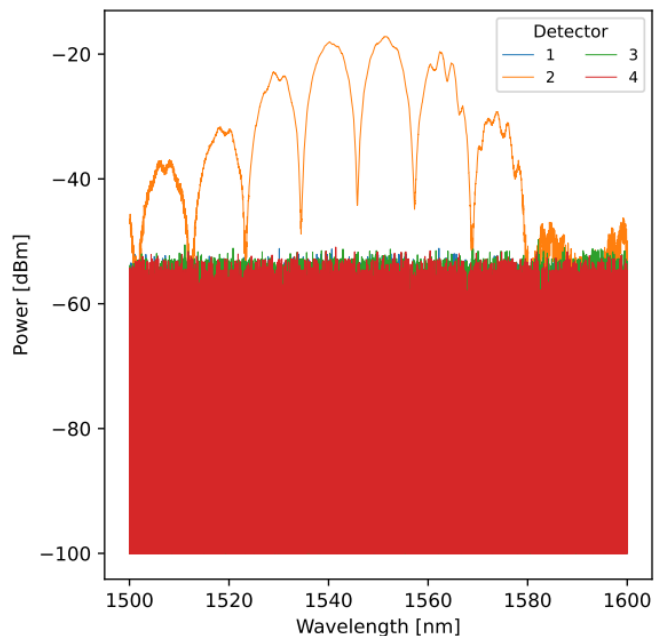


Fig. 14. Measured spectrum of MZI3 where $\Delta L = 50\mu m$

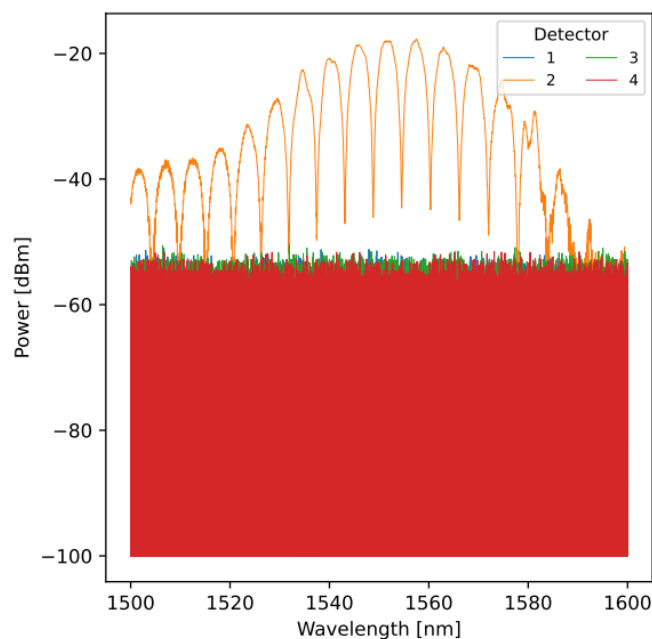


Fig. 13. Measured spectrum of MZI2 where $\Delta L = 100\mu m$

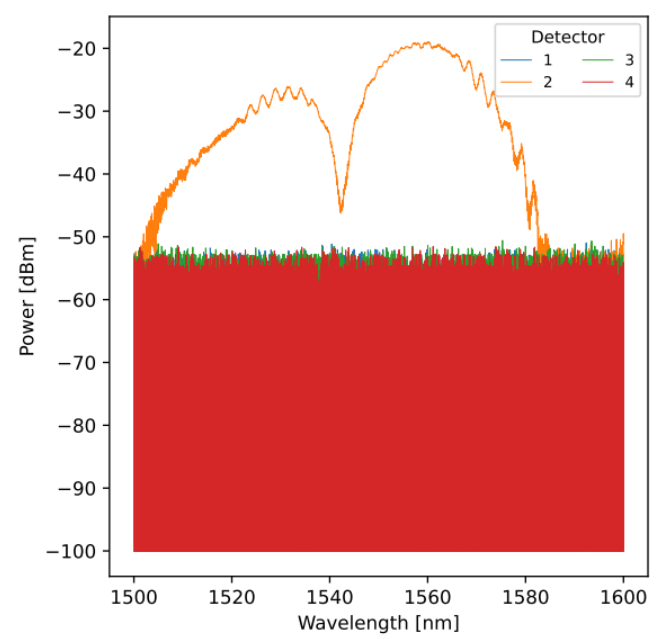


Fig. 15. Measured spectrum of MZI4 where $\Delta L = 13\mu m$

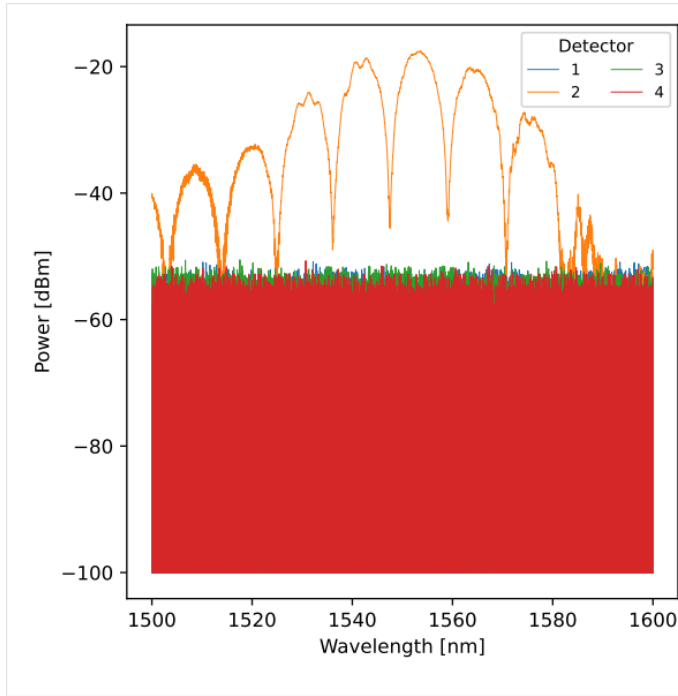


Fig. 16. Measured spectrum of MZI5 where $\Delta L = 50\mu\text{m}$ (testing the fabrication/measurement variations)

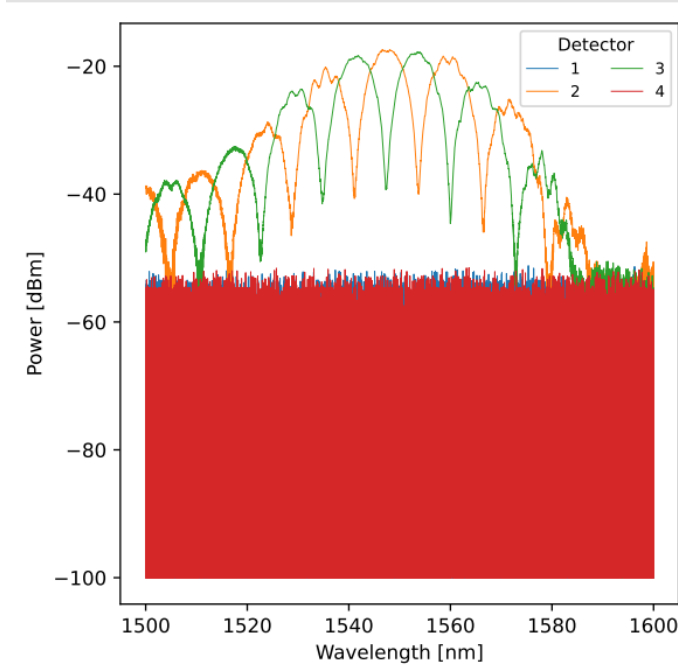


Fig. 17. Measured spectrum of MZI6

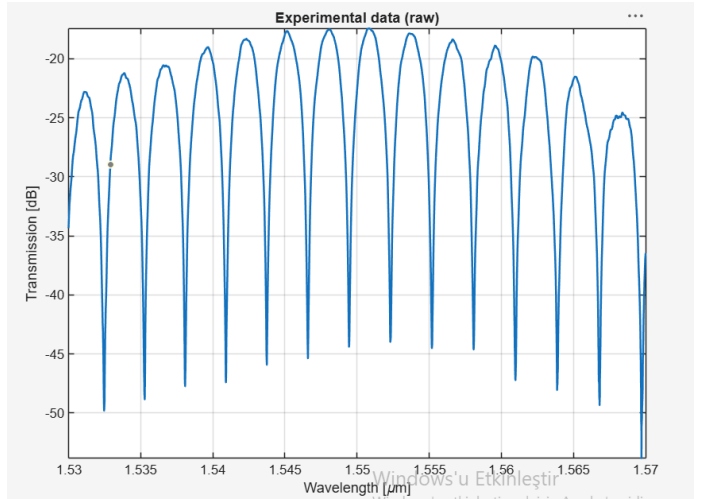


Fig. 18. Measured spectrum of MZI1 in MATLAB

The fourth-order polynomial provided a flatter baseline across the wavelength range while preserving the MZI interference fringes. Higher-order polynomials were avoided to prevent fitting the interference pattern itself.

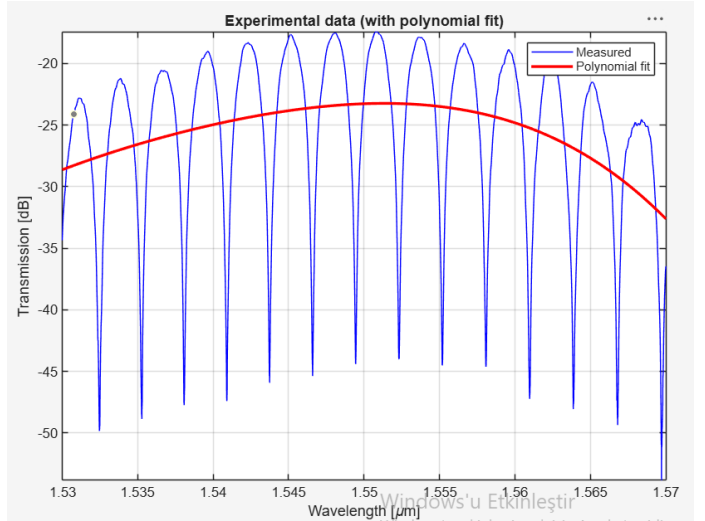


Fig. 19. 4th order polynomial fit for MZI1

Polynomial fit is subtracted from experimental data to get the baseline corrected version.

The baseline-corrected transmission spectrum of the MZI was fitted to an analytical transfer function using MATLAB's `lsqcurvefit` function. The model represents the effective index as a second-order Taylor expansion around the central wavelength and includes propagation loss as a fitting parameter. Initial parameter estimates were obtained from the measured FSR. The fitting converged successfully with a coefficient of determination of $R^2 = 0.988$, indicating excellent agreement between the measured spectrum and the analytical model. The fitted model was then used to extract the waveguide effective index, group index, and dispersion from the script.

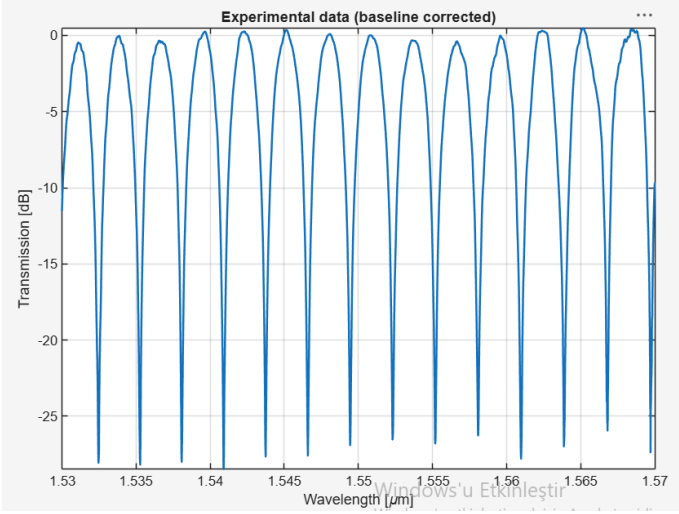


Fig. 20. Baseline correction for MZI1

The transmission spectrum of the Mach-Zehnder interferometer was modeled using the following transfer function:

$$F = 10 \log_{10} \left(\frac{1}{4} \left| 1 + \exp \left(-i \frac{2\pi n_{\text{eff}}}{\lambda} \Delta L - \frac{\alpha}{2} \Delta L \right) \right|^2 \right) + b \quad (13)$$

where n_{eff} is the effective index of the waveguide, λ is the wavelength, ΔL is the optical path length difference between the two interferometer arms, α is the propagation loss coefficient, and b is a constant offset accounting for the measured transmission level. The effective index was represented by a second-order Taylor expansion about the central wavelength to account for wavelength dependence and material dispersion. The model parameters were determined by nonlinear least-squares fitting using MATLAB's `lsqcurvefit` function.

As we can see, extracted group index for 1550 nm is roughly 4.196, which fits within the range determined by the corner analysis. From this we can also find FSR from eq. (9), as FSR=2.863 nm, which also fits within the range determined by the corner analysis.

VII. CONCLUSION AND ACKNOWLEDGEMENTS

A. Conclusion

A Mach-Zehnder interferometer (MZI) was designed, simulated, fabricated, and characterized using the silicon photonics design flow. Waveguide simulations in Lumerical MODE were used to determine the effective and group indices, while the MZI response was verified using Lumerical INTERCONNECT. Corner analysis demonstrated that expected fabrication variations produced only small changes in the effective index, group index, and free spectral range (FSR), indicating that the design is robust against normal process variations.

For the selected design MZI1, experimental measurements were analysed in MATLAB by first applying baseline correction to the measured transmission spectrum. The extracted group index at a wavelength of 1550 nm was found to be

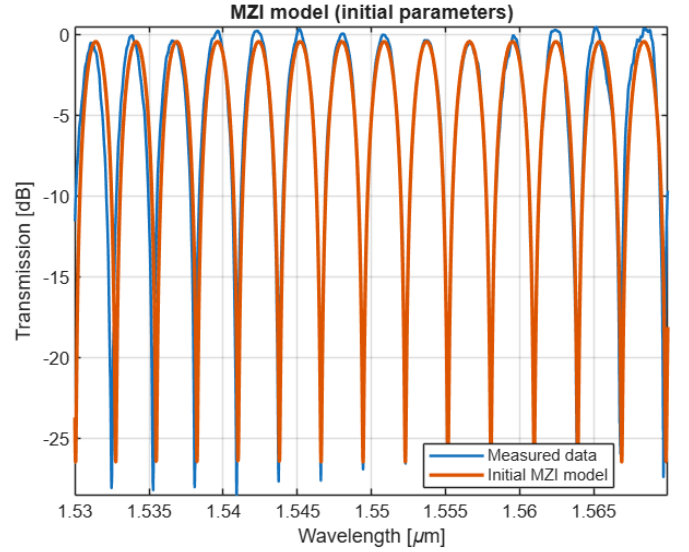


Fig. 21. Initial parameters versus measured results for MZI1

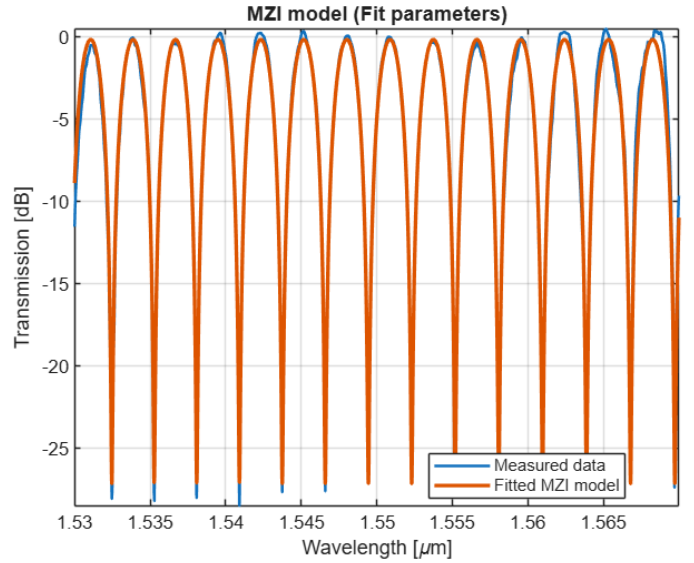


Fig. 22. Fitted model of MZI1

approximately 4.196, which agrees well with the simulated nominal value and lies within the range predicted by the corner analysis. Using the measured group index and a path length difference of 200 μm , the corresponding FSR was calculated to be approximately 2.863 nm, which is also consistent with both the simulation and theoretical prediction.

Overall, the measured device exhibited good agreement with the design and simulation results, demonstrating successful implementation and characterization of the silicon photonic Mach-Zehnder interferometer.

B. Acknowledgements

I acknowledge the edX UBCx Phot1x Silicon Photonics Design, Fabrication and Data Analysis course, which is supported

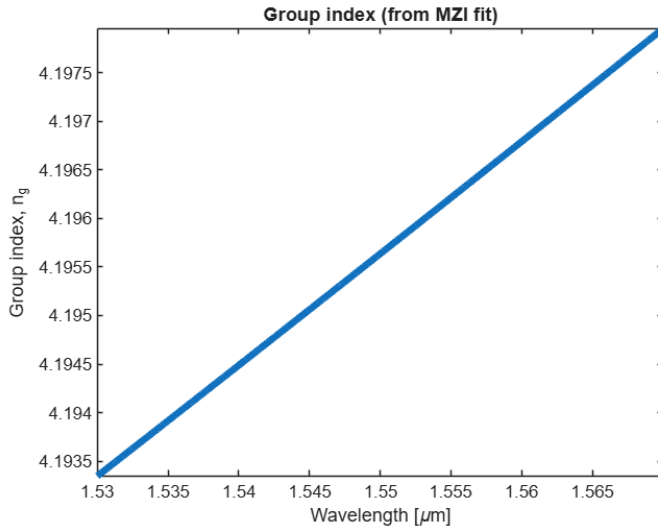


Fig. 23. Extracted group index versus wavelength plot for MZI1

by the Natural Sciences and Engineering Research Council of Canada (NSERC) Silicon Electronic-Photonic Integrated Circuits (SiEPIC) Program. The devices were fabricated by Richard Bojko at the University of Washington Washington Nanofabrication Facility, part of the National Science Foundation's National Nanotechnology Infrastructure Network (NNIN), and Cameron Horvath at Applied Nanotools, Inc. Mateo Branion-Calles performed the measurements at The University of British Columbia. We acknowledge Lumerical Solutions, Inc., Mathworks, Mentor Graphics, Python, and KLayout for the design software.

VIII. REFERENCES

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